BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD TESTING PROCEDURE FOR DARROW'S FULL STRENGTH SOLUTION (500ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPF/008**	**PAGE NO.:**	**Page 1 of 13**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2023/ IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**Description**	Clear, colourless solution, free from any visible particles. No leakage on the pouch when squeezed with hand.
2.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
3.	**Sodium Lactate (SL)**	1. Measure a volume of 50mL of product into a conical flask and evaporate to dryness. 2. Add 25mL of glacial acetic acid and allow to stir for 5min 3. Titrate with 0.1 M perchloric acid using crystal violet as indicator. 1 mL of 0.1 M perchloric acid is equivalent to 11.21 mg of $C_3H_5NaO_3$. $$\% \text{ sodium lactate} = \frac{\text{T X factor X } 11.21}{486} \text{ X } 100$$ Carry out the titration in duplicate.
4.	**Total Chloride (Cl^-)**	1. Measure 10mL of sample into a conical flask. 2. Add 1-2 drops 5% w/v Potassium Chromate indicator.

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		3. Titrate with silver nitrate solution until brick red colour endpoint. 1mL of 0.1M $AgNO_3$ is equivalent 3.55mg of total chloride. $\text{Chloride (\%)} = \frac{(T-Tb)\text{ X factor X } 3.55}{36.6}$X 100 Carry out the titration in duplicate.
5.	**BACTERIAL ENDOTOXIN**	**<u>PYROGEN TEST</u>** – Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed. – Withhold all food from the rabbits used during the period of the test – Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring. – All the rabbits which are to be used for the test should be

		more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit. – Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm. – If the initial temperature is less than 38.8^{0}c or more then 39.8^{o}cexclude the animal from the test. – Ensure all the needles and syringe to be used are new – Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c. – Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection. – Any rabbits showing more than 0.2^{0}c temperature

		difference between two successive readings in the last three readings can be excluded from the test. – Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes. – Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution. – If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected. – After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the

		injection. – The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution **<u>Limulus Amebocyte Lysate LAL Test</u>** - Open the seals of the test tubes of lyophilized LAL Reagents - Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes. - Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam. - Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation. - If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit. - Sample to be tested should be stored under conditions that

		bacteriological activities are controlled **Results** - Read reaction tubes one by one. - Invert the tube in one smooth motion. - A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. - A negative result is characterized by the absence of solid clot after inversion. **Testing Methodology of Bacterial Endotoxin (LAL) Test** - Each specimen shall be tested in duplicate - NMC: Take 100µl(0.1ml) LRW - PWC: Take 50µl(0.05ml) LRW and 50µl(0.05ml) CSE of 4λ - NPC: Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution. - PPC: Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

		- Add the LAL reagents as mentioned below in the table 1 - Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min. - After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations. - If a firm gel has formed, record the results as positive. - A result is negative, if an intact gel is not formed. Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material, WFI and finished product.
6.	**STERILITY TEST**	I. Sample for Finished Products: Randomly select 20units as per batch size for the sterility test. II. Transfer the samples after collection to QC department (microbiology laboratory) in clean plastic crates. III. Wipe the sample individually with 70% ethanol solution and

		keep in a clean stainless steel tray marked with Product Name, Batch No., and then transfer the samples to the sterility room for performing sterility. IV. Transfer the pre-incubated sterile media bottles, control media plates, sterile swabs, cotton wool, and micro-pipette to the sterility test room. V. Enter in sterility room as per the Entry / Exit procedure for Sterility Room. VI. Start the LAF as per SOP for operating Instruction for LAF (BML-QDG-MC-008). Disinfect using 70% ethanol solution. Wipe out all samples to be tested for sterility with 70% ethanol solution. VII. Check the Manometer reading of working LAF and check the temperature as well as humidity of the sterility room. Manometer reading of working LAF chamber pressure should be between 08 – 15mm of water. Temperature

		reading of the sterility room should be 27ºC ± 2ºC. VIII. Expose the control plates as specified locations throughout the testing. Open all open controls and don't open all closed controls. IX. Arrange the checked FTM and TSB media bottles in triplicates and label each according to the sample to be inoculated. X. Transfer aseptically 1ml of the product to be examined into TSB and FTM respectively. XI. Observe cultured bottles daily or at intervals during the incubation period for turbidity, using the Result Observation for sterility Test. XII. If the material being tested renders the medium turbid so that the presence or absence of microbial growth cannot be readily determined by visual examination, on the 14th day of incubation, transfer portions of the incubated medium

		(each not less than 1ml) of the medium to freshly prepared medium McCartney bottles (each not less than 9ml) of the same medium and then incubate the original and freshly prepared medium for not less than 4 days. XIII. If no evidence of microbial growth is found the product must be examined re-tested for verification with the test for sterility. XIV. If evidence of microbial growth is found after 4days incubation, the product to be examined does not comply with the test for sterility, unless it can be clearly demonstrated that the test was invalid for causes unrelated to the product to be examined.
7.	**MICROBIAL ENUMERATION TEST**	- Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method. - Testing to be performed under LAF of specified area

		- Arrange all the required material on the LAF before starting the test. - Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes - Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP (BML-QDG-MC-011) (cooled approximately up to 45°C) in two petri plates for TAMC - Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC - Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate - Simultaneously perform the negative control by plating

		1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45°c and allow for solidification. - Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20^{0}C to 25°C for 5 days in an inverted position.

*factor =standard factor of titrant

*T = Titre

*T_b = chloride correction titre = 0.2Ml

*486 = weight of sodium lactate analysed.

*11.21 = equivalent weight of sodium lactate.

*3.55 = equivalent weight of total chloride.

*18.3 = weight of total chloride analysed.

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*LAF – Laminar Airflow

*TSB – Tryptone soy broth

*FTM - Fluid Thioglycollate medium

*DHS - dry heat sterilization

*LAL ---- Limulus Amebocyte Lysate

*LRW --- LAL reagent water

*CSE --- Control sample Endotoxin

*PPC --- Positive product control

*cfu/ml --- Colony Forming Unit per Millilitre

*EU/ml--- Endotoxin Unit per Millilitre

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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